

Question One (8 marks)

- a) What is the common difference of the arithmetic series $2 + \frac{1}{2} - 1 + \dots$? **1**
- b) Evaluate $\sum_{k=3}^6 k^2$ **2**
- c) In a geometric series the 3rd term is 16 and the 7th term is 256. Find the common ratio of the series. **3**
- d) Find $f''(x)$ if $f(x) = 2x^4$ **2**

Question Two (8 marks)

- a) Simplify $\cos 15^\circ \sin 30^\circ - \sin 15^\circ \cos 30^\circ$ (Do not evaluate) **2**
- b) Prove the following identity: $\frac{2 \sin^2 x - 1}{\sin x + \cos x} \equiv \sin x - \cos x$ **3**
- c) Solve $3 \cos \theta + 4 \sin \theta = -3$ for $0^\circ \leq \theta \leq 360^\circ$ using the substitution $t = \tan \frac{\theta}{2}$. **3**

Question Three (8 marks)

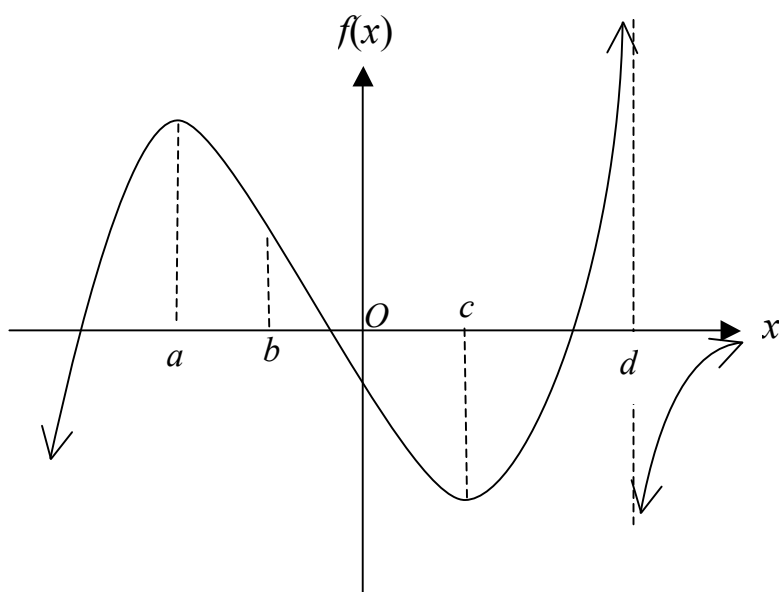
- a) Prove the following result using Mathematical Induction:
 $1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{1}{4}n^2(n+1)^2$ where n is a positive integer **4**
- b) L is the fixed point $(-1, 4)$ and P is a variable point on the parabola $x^2 = 8y$. M is the midpoint of LP . Find the equation of the locus of M . **4**

Question Four (9 marks)

a) Show that $\frac{x^2 + 9}{x - 4} = x + 4 + \frac{25}{x - 4}$ 1

b) Sketch $y = \frac{x^2 + 9}{x - 4}$ showing all turning points, intercepts and asymptotes.
(Do not find any points of inflexion) 6

c) Consider this graph, which has a point of inflexion at $x = b$.



For what values of x is $f'(x) > 0$? 2

Question Five (8 marks)

a) Consider the series: $(x + 3) - (x + 3)^2 + (x + 3)^3 - (x + 3)^4 + \dots$
i. For what values of x does this series have a limiting sum? 2

ii. Find the limiting sum in terms of x . 2

iii. Hence show that the limiting sum of this series cannot be greater than 1 1

b) The formula for the n th term of a series is $T_n = (n + 2)^3 - (n + 1)^3$.

i. Evaluate S_5 for this series. 2

ii. Find the sum of the first 100 terms of this series. 1

Question Six (9 marks)

a) Show by Mathematical Induction that $5n - 7 \leq 2^n$ where n is an integer greater than 2. **3**

b) $f(x)$ is the function $y = x^3 - ax^2 + b$, where a and b are positive.

i. Show that graph of $y = f(x)$ has a non-horizontal point of inflexion at $x = \frac{a}{3}$. **3**

ii. Given that $x_0 \leq \frac{2a}{3}$, show that $f(x_0) \leq b$. **3**

Question One (8 marks)

a) -1.5

b) $3^2 + 4^2 + 5^2 + 6^2 = 86$

c)

$ar^6 = 256 \quad (1)$

$ar^2 = 16 \quad (2)$

$(1) \div (2)$

$r^4 = 16$

$r = \pm 2$

d)

$f'(x) = 8x^3$

$f''(x) = 24x^2$

Question Two (8 marks)

a)

$\sin(30 - 15)$

$= \sin 15^\circ$

b)

$$\begin{aligned}
 lhs &= \frac{2 \sin^2 x - 1}{\sin x + \cos x} \\
 &= \frac{2 \sin^2 x - (\sin^2 x + \cos^2 x)}{\sin x + \cos x} \\
 &= \frac{\sin^2 x - \cos^2 x}{\sin x + \cos x} \\
 &= \frac{(\sin x + \cos x)(\sin x - \cos x)}{\sin x + \cos x} \\
 &= \sin x - \cos x \\
 &= rhs
 \end{aligned}$$

c)

$$\begin{aligned}
 \frac{3 \times (1 - t^2)}{1 + t^2} + \frac{4 \times 2t}{1 + t^2} &= -3 \\
 -3t^2 + 8t + 3 &= -3(1 + t^2) \\
 &= -3 - 3t^2 \\
 8t + 6 &= 0 \\
 t &= -\frac{3}{4} \\
 \tan \frac{\theta}{2} &= -\frac{3}{4} \\
 \frac{\theta}{2} &= 143.1301\dots, \quad 0 \leq \frac{\theta}{2} \leq 180 \\
 \theta &= 286^\circ 16'
 \end{aligned}$$

$Check \quad \theta = 180^\circ$

$lhs = 3 \cos(180^\circ) + 4 \sin(180^\circ)$

$= 3 \times -1 + 4 \times 0$

$= -3$

$= rhs$

$So \quad \theta = 180, 286^\circ 16'$

Question Three (7 marks)

a)

When $n=1$:

$lhs = 1^3$

$= 1$

$rhs = \left(\frac{1}{4} \times 1^2 \times (1+1)^2 \right)$

$= 1$

$= lhs$

So the statement is true when $n = 1$ Let k be a value of n for which the statement is true

Then

$1^3 + 2^3 + 3^3 + \dots + k^3 = \frac{1}{4} k^2 (k+1)^2$

We want to show that

$1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 = \frac{1}{4} (k+1)^2 (k+2)^2$

$lhs = \frac{1}{4} k^2 (k+1)^2 + (k+1)^3$

$= \frac{k^2 (k+1)^2}{4} + (k+1)^3$

$= (k+1)^2 \left[\frac{k^2}{4} + k+1 \right]$

$= (k+1)^2 \left(\frac{k^2 + 4k + 4}{4} \right)$

$= (k+1)^2 \left(\frac{(k+2)^2}{4} \right)$

$= \frac{1}{4} (k+1)^2 (k+2)^2$

$= rhs$

So if the statement is true when $n = k$ then it is also true when $n = k+1$.Since the statement is true when $n = 1$ it must also be true for $n = 2$, and since it is

true for $n = 2$, it must also be true for $n = 3$ and so on for all positive integers n .

b)
 $4a = 8$
 $a = 2$

So P has coordinates $(2ap, ap^2)$ i.e.
 $(4p, 2p^2)$

Let (X, Y) be the coordinates of M.
 Using the midpoint formula:

$$X = \frac{-1 + 4p}{2} \quad (1) \text{ and}$$

$$Y = \frac{4 + 2p^2}{2}$$

$$= 2 + p^2 \quad (2)$$

From (1) $p = \frac{2X + 1}{4}$

Into (2) :

$$Y = 2 + \left(\frac{2X + 1}{4} \right)^2$$

So the locus is the parabola

$$\left(x + \frac{1}{2} \right)^2 = 4(y - 2)$$

Question Four (6 marks)

a)
 i)

$$\frac{x^2 + 9}{x - 4} = \frac{x(x - 4) + 4(x - 4) + 25}{x - 4}$$

$$= x + 4 + \frac{25}{x - 4}$$

ii)

The vertical asymptote occurs at $x - 4 = 0$
 i.e. $x = 4$.

As $x \rightarrow \infty$, $\frac{25}{x - 4} \rightarrow 0$ so
 $y \rightarrow x + 4$

So $y = x + 4$ is an oblique asymptote

Since $x^2 + 9$ cannot equal zero there are no x -intercepts.

The y -intercept is $y = \frac{0^2 + 9}{0 - 4}$ i.e. $y = -\frac{9}{4}$

$$y' = \frac{(x - 4)(2x) - (x^2 + 9)(1)}{(x - 4)^2}$$

Stationary points will occur when

$$(x - 4)(2x) - (x^2 + 9)(1) = 0$$

$$x^2 - 8x - 9 = 0$$

$$(x - 4)^2 = 9 + 16$$

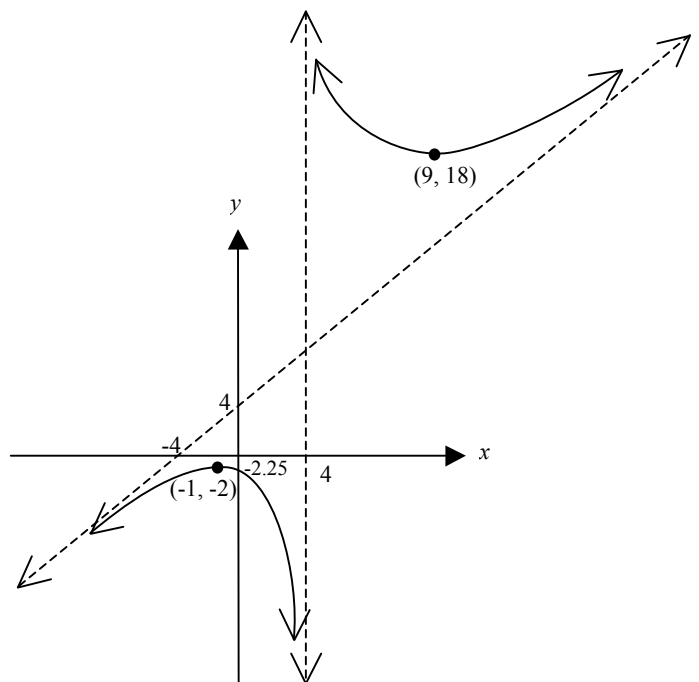
$$x = 4 \pm \sqrt{25}$$

$$x = -1, 9$$

x	-2	-1	0	9	10
y'	$\frac{11}{36}$	0	$-\frac{9}{16}$	0	$\frac{11}{36}$

There is a maximum turning point at $x = -1$ and a minimum turning point at $x = 9$.

When $x = -1$ $y = -2$ and when $x = 9$
 $y = 18$



b) $x < a$, $c < x < d$, $x > d$

Question Five (6 marks)

a)

i)

$$\begin{aligned}
 -1 &< -(x+3) < 1 \\
 -1 &< x+3 < 1 \\
 -4 &< x < -2
 \end{aligned}$$

ii)

$$\begin{aligned}
 \frac{a}{1-r} &= \frac{x+3}{1-(x+3)} \\
 &= \frac{x+3}{x+4}
 \end{aligned}$$

iii)

now, for all x

$$x+3 < x+4$$

We can divide both sides by $x+4$ since from part (i) $x+4 > 0$

$$\frac{x+3}{x+4} < 1$$

So the limiting sum cannot be greater than one.

Alternative proof:

Suppose the limiting sum were greater than one.

$$\text{Then } \frac{x+3}{x+4} > 1.$$

We can multiply both sides by $x+4$ since from part (i) $x+4 > 0$

$$x+3 > x+4$$

But this would mean that $3 > 4$ which is false. This means that our original assumption must be false so the limiting sum cannot be greater than one.

b)

$$\begin{aligned}
 S_5 &= (1+2)^3 - (1+1)^3 + (2+2)^3 - (2+1)^3 \\
 &\quad + (3+2)^3 - (3+1)^3 + (4+2)^3 - (4+1)^3 \\
 &\quad + (5+2)^3 - (5+1)^3 \\
 &= 3^3 - 2^3 + 4^3 - 3^3 + 5^3 - 4^3 + 6^3 - 5^3 \\
 &\quad + 7^3 - 6^3 \\
 &= 335
 \end{aligned}$$

$$\begin{aligned}
 S_{100} &= (1+2)^3 - (1+1)^3 + (2+2)^3 - (2+1)^3 \\
 &\quad + (3+2)^3 - (3+1)^3 \dots + (100+2)^3 \\
 &\quad - (100+1)^3 \\
 &= 3^3 - 2^3 + 4^3 - 3^3 + 5^3 \\
 &\quad - 4^3 + \dots + 102^3 - 101^3
 \end{aligned}$$

Almost every term in the sum cancels so

$$\begin{aligned}
 S_{100} &= 102^3 - 2^3 \\
 &= 1061200
 \end{aligned}$$

Question Six (7 marks)

a)

$$\begin{aligned}
 \text{Let } n &= 3 \\
 lhs &= 5(3) - 7 \\
 &= 8 \\
 rhs &= 2^3 \\
 &= 8 \\
 &\geq lhs
 \end{aligned}$$

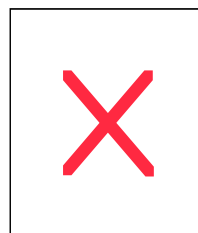
Let k be a value of n for which the statement is true

$$\text{i.e. } 5k - 7 \leq 2^n$$

$$\text{we want to show that } 5(k+1) - 7 \leq 2^{k+1}$$

note that since $k \geq 3$,

$$\begin{aligned}
 2^k &\geq 2^3 \\
 &= 8
 \end{aligned}$$



So if the statement is true when $n = k$ then it is also true when $n = k + 1$.

Since the statement is true when $n = 3$ it must also be true for $n = 4$, and since it is true for $n = 4$, it must also be true for $n = 5$ and so on for all positive integers n .

b)

i.

$$f(x) = x^3 - ax^2 + b$$

$$f'(x) = 3x^2 - 2ax$$

$$f''(x) = 6x - 2a$$

$$\text{When } x = \frac{a}{3},$$

$$f''(x) = 6\left(\frac{a}{3}\right) - 2a \\ = 0$$

$$f'(x) = 3\left(\frac{a}{3}\right)^2 - 2a\left(\frac{a}{3}\right) \\ = -\frac{a^2}{3} \\ < 0, \text{ since } a > 0$$

$$\text{When } x < \frac{a}{3}$$

$$6x < \frac{6a}{3}$$

$$\text{so } 6x - 2a < \frac{6a}{3} - 2a \\ = 0$$

$$\text{i.e. } f''(x) < 0$$

When



$$6x > \frac{6a}{3}$$

$$\text{so } 6x - 2a > \frac{6a}{3} - 2a \\ = 0$$

$$\text{i.e. } f''(x) > 0$$

So when $x = \frac{a}{3}$, the second derivative equals zero and changes signs and the first derivative is not zero i.e. there is a non-horizontal point of inflexion at $x = \frac{a}{3}$.

ii) Note that since $f(x)$ is a polynomial, the graph of $y = f(x)$ is continuous.

Stationary points occur when $f'(x) = 0$

$$3x^2 - 2ax = 0$$

$$x(3x - 2a) = 0$$

$$x = 0, \frac{2a}{3}$$

$$\text{When } x = 0$$

$$f''(x) = 6(0) - 2a \\ = -2a \\ < 0 \text{ since } a > 0$$

So there is a maximum turning point at $(0, b)$

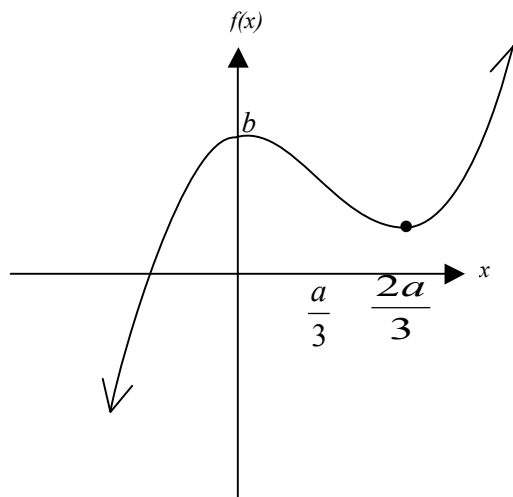
$$\text{When } x = \frac{2a}{3},$$

$$f''(x) = 6\left(\frac{2a}{3}\right) - 2a \\ = 2a \\ > 0 \text{ since } a > 0$$

So there is minimum turning point at

$$x = \frac{2a}{3}$$

The location of these points depends on a and b but we can conclude that the *shape* of the graph is:



So when $x_0 \leq \frac{2a}{3}$, show that $f(x_0) \leq b$.